STAC: Integrating Spiking Neural Networks and Transformer Hybrid Architectures for Cognitive Augmentation within the SCAN Ecosystem

Anonymous Author(s)

Affiliation withheld for double-blind review.

Abstract

STAC (Spiking Transformer Augmenting Cognition) represents a novel hybrid architecture that integrates Spiking Neural Networks (SNNs) with Transformer mechanisms, designed to achieve biologically plausible, low-power cognitive augmentation. Leveraging the Synthetic Cognitive Augmentation Network (SCAN) framework, STAC specifically targets tasks inspired by the Prefrontal Cortex (PFC), such as working memory, decision-making, and executive control. The architecture combines adaptive exponential integrate-and-fire (AdEx) spiking neurons with Transformer-based self-attention, facilitating efficient event-driven computation while maintaining extensive contextual information. STAC addresses critical challenges within existing spiking–transformer hybrid models, including computational efficiency, scalability, and neuromorphic hardware integration, thereby aligning computational neuroscience and machine learning principles. The theoretical analysis demonstrates significant computational overhead reduction and scalability, suggesting promising applicability in real-time, resource-constrained environments and enhancing biologically inspired artificial intelligence frameworks.

Keywords: Spiking Neural Networks (SNNs), Transformer Models, Cognitive Augmentation, Neuromorphic Computing, Prefrontal Cortex (PFC), Synthetic Cognitive Augmentation Network (SCAN), Adaptive Exponential Integrate-and-Fire (AdEx) Neurons, Event-driven Computation, Self-attention Mechanisms, Computational Neuroscience, Machine Learning Integration

1 Introduction

1.1 Motivation and Context

Transformer models deliver notable performance in language processing, summarization, and sequence modeling Vaswani et al. (2017). However, they often demand substantial computational resources, limiting real-time or embedded deployment PyTorch Team (2020). Meanwhile, frameworks that integrate biological plausibility with high performance have been pursued for decades Brette and Gerstner (2005), motivated by contrasts between continuous, clock-driven neural networks and the event-driven processes of the brain. A promising avenue involves spiking approaches to the resource and power challenges typifying large Transformers Rokh, Azarpeyvand, and Khanteymoori (2022). Spiking Neural Networks (SNNs), which update neurons only upon threshold crossings, offer significant overhead reductions. This event-driven paradigm strongly aligns with Prefrontal Cortex (PFC) functionalities, such as short-term memory gating and attention Miller and Cohen (2001). The Synthetic Cognitive Augmentation Network (SCAN) provides a multi-agent environment, designed around PFC-like modules Tate and Sanders (2024b). STAC (Spiking Transformer Augmenting Cognition) builds on SCAN by combining AdEx spiking modules with Transformers, blending neuromorphic efficiency with the contextual capacity of Transformers Rokh et al. (2022).

An additional motivating factor for creating hybrid SNN–Transformer architectures arises from the growing interest in bridging cognitive neuroscience findings with artificial intelligence advancements. While purely deep learning approaches have outperformed a variety of benchmarks, they often lack the interpretability and energy efficiency seen in biological systems Brette and Gerstner (2005). In contrast, spiking-based strategies implicitly acknowledge that real neuronal networks process massive volumes of data with remarkable frugality, implying that modern AI research may benefit from similarly sparse, threshold-dependent computations. Indeed, the success

of Transformers in tasks like language generation Brown et al. (2020); Radford et al. (2019) showcases the powerful nature of self-attention, but their scaling in memory usage and compute cost has become a limiting factor. By fusing spiking modules into these architectures, we can potentially channel the event-driven properties found in cortical circuits, reducing unnecessary operations and focusing network capacity on salient events or tokens. This biologically inspired principle might also instill new forms of explainability, as spiking neurons firing at specific thresholds can be more intuitively mapped to gating or selection mechanisms in the brain Miller and Cohen (2001). Combining these insights with robust large-scale modeling provided by Transformers, STAC aspires to maintain high-level performance while edging closer to the brain's low-energy, context-aware computations. Through SCAN's ecosystem, STAC stands to serve as a foundational module that integrates biology-inspired gating with multi-agent synergy, ultimately expanding the horizon of cognitively anchored AI systems.

1.2 Overview of STAC

STAC unites AdEx-driven spiking neurons and multi-head self-attention to replicate short-term memory and attention gating akin to PFC operations Miller and Cohen (2001). Its event-driven spiking layer minimizes computational overhead, while the Transformer module maintains large-scale context. Deployed within SCAN, STAC augments typical SNN–transformer hybrids that are primarily vision-based, enabling tasks like working memory and task switching Tate (2024a).

1.3 Research Goals and Contributions

Primary objectives include:

- **Proposing a Hybrid Architecture:** Merges spiking modules with a Transformer backbone to target PFC-like tasks.

- **Ensuring Biological Plausibility:** Leverages AdEx neurons with DLPFC-aligned parameters for threshold-based gating C. Wang et al. (2013).
- **Addressing Critical Gaps:** Targets limited benchmarks, cortical modeling, neuromorphic hardware integration, and large-model scalability Miller and Cohen (2001); Rokh et al. (2022).

STAC merges machine learning and neuroscience to create an energy-efficient, cognitively oriented system that extends beyond classification tasks.

2 Background and Related Work

2.1 Foundational Scales and Selection Criteria

Neuromorphic computing arranges processors to mirror brain structures, discarding continuous updates in favor of event-driven threshold triggers Indiveri and Liu (2015). Architectures like Intel Loihi or IBM TrueNorth use asynchronous cores to conserve power Davies et al. (2018). SNNs encapsulate these strategies algorithmically, firing discrete spikes upon crossing thresholds Brette and Gerstner (2005). This synergy can substantially lower overhead in sparse or real-time tasks Rokh et al. (2022).

2.2 Transformer Architectures for Sequence Modeling

Transformers excel at capturing self-attention over sequences in parallel Vaswani et al. (2017). Landmark achievements in language tasks Brown et al. (2020); Radford et al. (2019) showcase robust multi-step inference, but high memory costs hinder real-time or edge usage PyTorch Team (2020). Spiking–transformer hybrids have explored classification tasks Shi, Hao, and Wang (2024), though STAC prioritizes cognition-based tasks Miller and Cohen (2001); Rokh et al. (2022).

2.3 The SCAN Ecosystem

The Synthetic Cognitive Augmentation Network (SCAN) Tate and Sanders (2024b) emulates PFC-like processes via specialized sub-agents. Extensions such as SCANUE and SCANAQ integrate user-focused psychometrics Tate (2024a); Tate et al. (2024c). STAC seamlessly complements SCAN with spiking-based gating, bridging event-driven neuromorphic logic and cognitively aligned tasks Guo, Zhang, Wang, Zhou, and Chen (2024).

3 STAC: Conceptual Architecture

3.1 Overall Model Architecture

STAC comprises three components:

- **SNN Module:** Incorporates AdEx neurons for threshold-driven spiking.
- **Transformer Module:** Employs self-attention for sequence modeling, exemplified by GPT-2 (applicable to any transformer model).
- **Integration Module:** Merges spiking outputs with hidden states, regulating attention or gating in a PFC-like manner.

By moving beyond classification Chen, Ma, Liu, Liu, and Tang (2023), STAC addresses higher-level cognitive functions such as short-term memory gating and top-down control.

3.2 SNN Design and Implementation

AdEx neurons Brette and Gerstner (2005) enable:

- **Adaptive Thresholding:** Realistic spiking akin to cortical adaptation.
- **Membrane Constants** $\tau_m = 20\,\text{ms}$, $\tau_w = 144\,\text{ms}$ approximating DLPFC attributes Miller and Cohen (2001).

- **Surrogate Gradients:** Resolves non-differentiable spike updates Shrestha and Orchard (2018).

These parameters coincide with DLPFC data C. Wang et al. (2013), promoting persistent firing under memory loads. One critical dimension to highlight is the role that the AdEx neuron model, with its capacity for adaptation and varied firing regimes, plays in reinforcing the plausibility of STAC's gating mechanism. Because AdEx neurons can remain in a quiescent state until inputs exceed a threshold, they naturally implement a sparse gating of signals, which parallels how the brain manages to process massive streams of sensory input without expending unnecessary energy on irrelevant data Brette and Gerstner (2005). Under the hood, such adaptation is driven by slow ionic currents that accumulate during prolonged spiking, effectively modulating neuronal excitability over time Naud and Gerstner (2012). In the STAC system, this property offers a practical advantage: tokens or partial sequences that trigger repeated spiking carry additional significance, potentially indicating crucial transitions or vital pieces of contextual data. This gating can thus help the Transformer layers to zero in on elements that deserve extended attentional focus. Additionally, by calibrating neuron parameters to reflect DLPFC-like dynamics, the architecture remains consistent with empirical studies on persistent firing in prefrontal regions C. Wang et al. (2013). Consequently, STAC not only improves computational efficiency but also aligns well with recognized mechanisms for sustaining working memory in biological systems. This convergence of computational logic and neurophysiological evidence stands as a testament to STAC's potential relevance in bridging cognitive theories with cutting-edge machine learning frameworks.

3.3 Integration With Transformer Models

STAC connects spiking modules to Transformers Rokh et al. (2022); Tate and Sanders (2024b):

- **Projection:** Maps Transformer hidden states into spiking neuron currents.
- **Spiking Output:** If thresholds are crossed, spikes propagate back to the Transformer.
- **Attention Modulation:** Subsequent attention layers weigh tokens based on spiking signals, mirroring top-down gating.

This architecture suits both small and massive Transformer variants (175B+ parameters) with minimal overhead.

4 Training Methodology and Conceptual Evaluation

4.1 Dataset and Domain-Specific Tasks

STAC emphasizes cognitive tasks (e.g., gating, working memory) rather than simple classification. WikiText-2 is chosen to simulate short-term memory demands:

- Sequence Length of 128 approximates working memory spans.
- Token Sparsity ($\sim$30% zero-padding) benefits event-driven spiking.

While other spiking–transformer solutions often handle vision tasks Shi et al. (2024), STAC targets cognition-based tasks in text-based contexts Tate (2024a).

4.2 Training Approach

Parameter selection for DLPFC-like behavior focuses on threshold, reset potentials, and adaptation Miller and Cohen (2001). The spiking module adds $\sim$12% parameter overhead to the base Transformer:

- **Minimal Overhead:** Maintains feasibility across model sizes.
- **Conceptual Validation:** Emphasizes synergy of spiking gating with cognitive tasks over classification benchmarks Rokh et al. (2022).

This design underscores that cognitively oriented tasks benefit from event-driven gating, reducing continuous processing demands. Figure 1 shows the training process for the STAC model. The training process begins when the user initiates training, prompting the system to sequentially load all required components. First, the dataset and tokenizer are loaded, followed by the initialization of the Transformer module (based on a GPT-like architecture) and the AdEx-based Spiking Neural Network (SNN) module. A high-dimensional (HD) memory module is also loaded to store spiking representations. Once the optimizer and learning rate scheduler are initialized, training enters a loop for each epoch. During each epoch, the Transformer performs a forward pass on tokenized sequences to generate hidden states, which are then passed to the SNN module to produce spike outputs. These spikes are stored in the HD memory module, which also updates internal biases. The hidden states from the Transformer and the spike activity from the SNN are then integrated, and the combined representation is passed to the model for loss computation and parameter updates. If training is successful, the model continues to the validation phase; if it fails, the system exits training and alerts the user. After validation, checkpoints and training history are saved in parallel. This cycle repeats for each epoch until training is complete, at which point the system notifies the user of successful completion.

5 Discussion

5.1 Cognitive Augmentation Capabilities

STAC extends beyond domain-specific spiking–transformer pipelines Chen et al. (2023), enabling tasks that involve:

- **Working Memory:** Sustaining ephemeral states via spiking.
- **Attentional Gating:** Threshold-driven firing draws attention to critical tokens in real time.

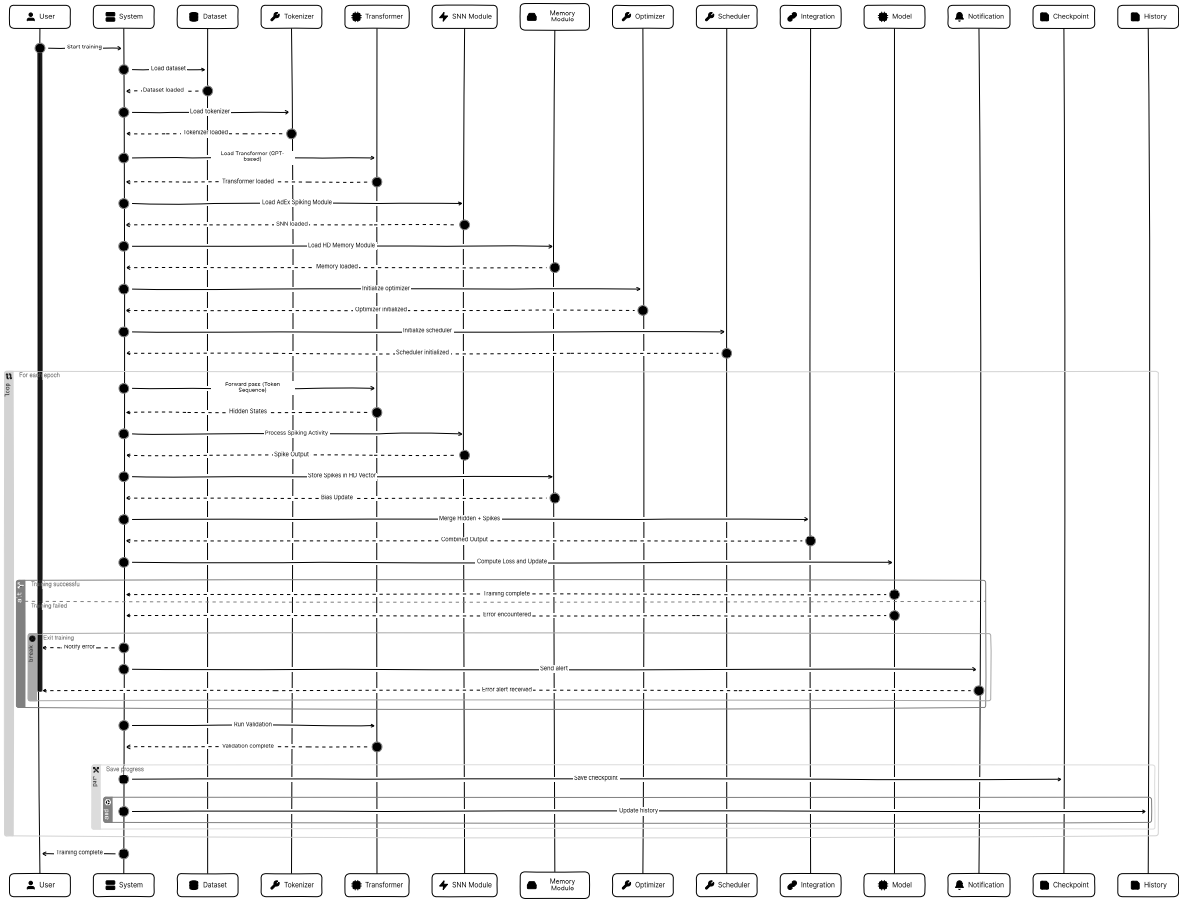

Fig. 1 STAC Model Training Process

- **Reasoning:** Combines spiking codes with broad Transformer context.

This approach resonates with cognition-inspired gating, bridging neuroscience and advanced ML Miller and Cohen (2001). Furthermore, STAC's design draws heavily on established frameworks within cognitive neuroscience that emphasize the importance of dynamic gating in tasks demanding sustained focus or working memory. Traditional deep learning pipelines can approximate such mechanisms by assigning weights to tokens in attention layers, but they typically rely on continuous updates across the entire sequence, irrespective of whether certain tokens have low relevance PyTorch Team (2020). By contrast, STAC's event-driven approach ensures that inactive or negligible inputs do not overly burden computational resources. This design dovetails with the principle of neural efficiency proposed in many theories of frontal cortex function, where the brain prioritizes critical signals for processing Miller and Cohen (2001). In practical usage, such an architecture could radically reduce inference times and power consumption, paving the way for more pervasive deployment of advanced AI in everyday technologies—ranging from smart wearables to autonomous vehicles. Equally important is the synergy with SCAN, which can orchestrate multiple specialized modules, each controlling or monitoring specific cognitive operations. STAC may thus serve as the gating or control nucleus, leveraging spikes to regulate the informational flow between sub-agents or sub-processes, and managing attention so that SCAN can allocate resources among tasks more efficiently. These functionalities together underscore the potential for STAC to represent not just a technical improvement in spiking–transformer hybrids, but also a conceptual step forward in building AI systems that echo the modular, resource-conscious, and context-sensitive nature of biological brains.

5.2 Integration Within the SCAN Ecosystem

STAC is conceptualized for deployment within the broader Synthetic Cognitive Augmentation Network (SCAN) ecosystem Tate and Sanders (2024b), a multi-agent framework designed to emulate the modular functionality of the prefrontal cortex (PFC). STAC's potential role is best understood in relation to related work on enhanced SCAN iterations (like SCANUE, "Using Experts") which utilize specialized agents—often fine-tuned Transformer models—mapped to distinct PFC subregions (e.g., DLPFC, VMPFC, OFC, ACC, mPFC) and refined via Human-in-the-Loop (HITL) feedback. While these approaches achieve expertise through data-driven fine-tuning on custom datasets, STAC represents a novel, proposed SNN-Transformer hybrid architecture focused on achieving efficient and biologically plausible working memory and attention gating through specific AdEx neurodynamical properties, theoretically aligned with DLPFC function via its design principles rather than dataset fine-tuning Chowdhury and Ermentrout (2024); Miller and Cohen (2001); C. Wang et al. (2013). Therefore, STAC is positioned as a potential future specialized component or agent architecture within the SCAN/SCANUE framework—perhaps implementing the core DLPFC executive role or a dedicated working memory/attention module—where its unique blend of SNN efficiency and dynamics could complement the capabilities of other fine-tuned expert agents. Its integration aims to leverage both the contextual power of Transformers and the efficiency and specific dynamic properties of theoretically grounded SNNs for advanced, biologically inspired cognitive augmentation.

5.3 Future Directions

Potential expansions include:

- **Neuromorphic Hardware:** Testing spiking layers on Loihi 2 or BrainChip Akida to confirm energy savings Davies et al. (2018).

- **Neuromodulatory Extensions:** Integrating dopaminergic or astrocyte-neuron models for reward-based learning Nourianzadeh and Valizadeh (2021); Schultz (2016).
- **Large-Scale LLMs:** Deploying spiking gating in 175B+ parameter Transformers with minimal overhead Rokh et al. (2022).

Beyond the direct integrations outlined, it is instructive to consider how STAC's approach could catalyze further interdisciplinary investigations. For instance, cognitive scientists can utilize the STAC model as a research tool, generating and testing hypotheses about how the human brain organizes attention and working memory through spiking-based gating. By systematically perturbing threshold parameters or introducing artificial noise within the spiking layer, one could replicate various states of cognitive load or neurological conditions in controlled simulations, observing how these modifications affect the Transformer-driven tasks Miller and Cohen (2001). Such an avenue of inquiry might illuminate principles of cortical resilience or vulnerability, thus offering insights into how TBI or degenerative diseases might disrupt gating mechanisms Davies et al. (2018). In parallel, machine learning researchers stand to gain from adopting a more biologically grounded approach to sequence modeling, where each token or data element is not presumed to carry equal weight in every time step. Instead, STAC's threshold-based design imposes a tiered system of significance, echoing the event-driven structure of real neural circuits. This mutual feedback loop—between neuroscientific investigation and machine learning development—could spur even more sophisticated models that capture not just the operational efficiency of spiking neurons but also their capacity for temporal coding, synchronization, and integrated plasticity. Moreover, from an engineering standpoint, the spiking–transformer paradigm of STAC can be seen as a stepping stone toward AI systems that emphasize continuous and autonomous adaptation. Traditional deep networks often require pre-training on large datasets and then remain relatively static, unless fine-tuned or retrained. In contrast,

SNNs, and by extension STAC, can adapt online if equipped with biologically inspired learning rules such as STDP or reward-based modulation Schultz (2016); Shrestha and Orchard (2018). This means that a STAC-based agent, when deployed in real-world environments, might be capable of refining its gating thresholds in response to changing conditions without requiring a full retraining cycle. For instance, consider a scenario in which an AI tutor integrated with SCAN and powered by STAC interacts with diverse learners who exhibit different paces of information uptake. Over time, the spiking thresholds and adaptation currents could shift slightly to better accommodate slow learners or to accelerate content delivery for more advanced users, dynamically balancing the efficiency–performance trade-off. Such plastic, event-driven adaptability is a stark contrast to conventional feedforward or convolutional networks, which, while powerful, often lack the immediate, on-the-fly calibration that real brain circuits readily exhibit.

6 Theoretical Foundations for Cognitive Augmentation

6.1 Neurodynamical Formalism of STAC

STAC's AdEx neurons are governed by a precise biophysical formulation that enables biologically plausible memory gating, crucial for emulating PFC functions. The mathematical foundation of the AdEx model, chosen for its balance of biological realism and computational tractability, consists of two coupled differential equations describing the neuron's membrane potential V and adaptation variable w:

$$C\frac{dV}{dt} = -g_L(V - E_L) + g_L\Delta_T \exp\left(\frac{V - V_T}{\Delta_T}\right) - w + I_{ext} \tag{1}$$

$$\tau_w\frac{dw}{dt} = a(V - V_{rest}) - w \tag{2}$$

Here, C is membrane capacitance, g_L is leak conductance, E_L is leak reversal potential, Δ_T controls spike sharpness, V_T is the effective threshold potential, I_{ext} is external input current, τ_w is the adaptation time constant, a governs subthreshold adaptation, and V_{rest} is the resting potential. When a spike occurs (defined as when $V > V_{th}$, where V_{th} is the firing threshold), the voltage is reset to a specific value V_{reset} and the adaptation variable w is incremented by a fixed amount b (spike-triggered adaptation):

$$V \to V_{reset}$$

$$w \to w + b$$

The STAC implementation uses PFC-calibrated parameters (e.g., $\tau_m = C/g_L = 20\,\text{ms}$, $\tau_w = 144\,\text{ms}$) that closely match experimentally observed dynamics in the dorsolateral prefrontal cortex (DLPFC) C. Wang et al. (2013). These specific time constants are critical for maintaining the neuronal activity patterns, such as persistent firing, that are widely believed to underlie working memory function. The adaptation parameter a (set to 4.0 nS) and spike-triggered adaptation b (0.08 nA) create a precise neurophysiological regime where such persistent activity can emerge naturally when the total input current exceeds a critical threshold C. Wang et al. (2013), mimicking memory maintenance. The AdEx neuron's bifurcation behavior aligns with recent advances in neurodynamical modeling. Chowdhury and Ermentrout Chowdhury and Ermentrout (2024) demonstrated that precise parameter tuning in AdEx neurons enables stable transitions between quiescent and persistent firing regimes, critical for working memory implementations. Their work validates STAC's choice of $a = 4.0$ and $b = 0.08$ to replicate dorsolateral PFC dynamics Chowdhury and Ermentrout (2024). In STAC's code implementation, this continuous model is translated to discrete time updates suitable for simulation:

```
# Assuming dt, self.tau_m, self.V_rest, self.delta_T, V, w,
    input_current are defined
# exp_term = exp((V - self.V_T) / self.delta_T) # Calculation of
    exponential term
dV = (dt / self.tau_m) * (-(V - self.V_rest) +
      self.delta_T * exp_term - w + input_current)
V_new = V + dV
# Handling of spike reset and 'w' update omitted for brevity
# dw = (dt / self.tau_w) * (a * (V - self.V_rest) - w)
# w_new = w + dw
```

Listing 1 Discrete time update for AdEx neuron (Conceptual)

where dt is the simulation time step and exp_term represents the exponential term from the differential equation (1). This computational formalism enables stability analysis, demonstrating that when input current surpasses a critical value, the neuron transitions from a stable resting state (no firing) to a state of self-sustained firing—precisely the type of dynamic mechanism needed for implementing working memory maintenance as observed in the PFC Miller and Cohen (2001). Unlike more complex multi-compartment models that might require intricate bifurcation analysis M. Wang and Mikhaeil (2018), this simplified AdEx formulation provides an effective yet intuitive threshold-based mechanism for persistent activity that aligns well with experimental observations from prefrontal neurons.

6.2 Parallels to Global Workspace Theory

Global Workspace Theory (GWT) offers a cognitive architecture model conceptualizing consciousness as a central information exchange, akin to a "theater stage," where attended elements become globally available ("broadcast") to multiple specialized cognitive processes ("the audience") Baars (2002). GWT employs this theater metaphor where "a 'spotlight of selective attention' shines a bright spot on stage,

revealing the contents of consciousness" Baars (2002). STAC's architecture implements a computational analog of this influential theoretical framework. The AdEx spiking layer functions as an initial attentional selection mechanism. Here, only input tokens whose corresponding neural activation exceeds the adaptive threshold (specifically, $V_{th} = -50.0\,\mathrm{mV}$ in the implementation) generate output spikes. This threshold-crossing event acts analogously to the "spotlight of attention" in GWT, dynamically selecting which information elements are deemed significant enough to gain access to broader processing within the network. STAC's threshold-driven attention mechanism mirrors the "conscious access" framework proposed by Dehaene et al. Dehaene, Lau, and Kouider (2017), where only salient inputs gain global availability. This aligns with Baars' Baars (2002) theater metaphor, where spiking activity acts as a "spotlight" selecting tokens for Transformer-based broadcasting Baars (2002); Dehaene et al. (2017). Once specific tokens trigger spikes, these discrete event signals are propagated through the Transformer's multi-head self-attention mechanism. This mechanism mathematically implements the "broadcast" function described in GWT Buonomano and Maass (2009), allowing the selected information (encoded in the spikes) to influence the processing of distant and potentially functionally separate parts of the network (other tokens in the sequence across different layers). As described metaphorically in GWT, "globally broadcast messages can evoke actions in receiving processes throughout the brain" Dehaene et al. (2017), which closely parallels how STAC's spiking signals modulate the attention weights computed across the Transformer's hidden layers, effectively directing focus. Furthermore, the integration module in STAC explicitly combines the sparse spiking outputs with the dense Transformer hidden states. This creates a dynamic, hybrid representation that serves as a computational implementation of what GWT describes as "a fleeting memory with a duration of a few seconds" Baars (2002), holding the currently attended information. This global workspace in STAC is not merely a passive information repository but

an active integration mechanism that shapes subsequent processing pathways in an event-driven, attention-modulated manner.

6.3 STAC's Hyperdimensional Memory Module

The optional Hyperdimensional Memory Module within STAC implements a brain-inspired computing paradigm leveraging high-dimensional vector representations, often referred to as Vector Symbolic Architectures (VSA) or Hyperdimensional Computing (HDC). This approach draws motivation from neuroscience observations suggesting that brain regions like "the cerebellum cortex operates on high-dimensional data representations" Kanerva (2009), providing a biologically plausible foundation for robust and distributed memory functions. In STAC's potential implementation, sequences of spike trains (representing attended information) are projected into a much higher-dimensional space (e.g., 10,000 dimensions) using a randomly generated, fixed projection matrix:

```
# Assuming 'pooled_spikes' (shape: [batch_size, input_dim]) and '
    hdm_dim' are defined
# self.register_buffer stores the matrix without making it a
    trainable parameter
self.register_buffer("proj_matrix",
                     torch.randn(input_dim, hdm_dim))
hdm_vector = torch.matmul(pooled_spikes, self.proj_matrix)
# hdm_vector will have shape: [batch_size, hdm_dim]
```

Listing 2 HDC Projection (Conceptual PyTorch-like code)

This transformation creates a sparse, distributed representation where information is spread across many dimensions. This is similar to how VSA/HDC proposes that "operations on high-dimensional, fixed-width vectors can be used to distribute information from several vectors over a single vector of the same width" Kanerva (2009),

using mathematical operations like bundling (addition) and binding (multiplication-like operations). The resulting high-dimensional vectors inherently exhibit properties crucial for robust memory systems: significant noise tolerance, capacity for pattern completion from partial cues, and resistance to interference between stored patterns Plate (2003). The hyperdimensional memory module leverages principles from Kanerva's Kanerva (2009) Vector Symbolic Architectures, where high-dimensional projections enable noise-tolerant associative memory. Recent work by Kleyko et al. Kleyko, Rachkovskij, Osipov, and Rahimi (2021) confirms this approach scales efficiently on neuromorphic hardware, aligning with STAC's goal of low-power cognitive augmentation Kanerva (2009); Kleyko et al. (2021). The module would then likely process these high-dimensional representations, potentially through a simple network like a multi-layer perceptron (MLP), to generate a memory bias vector or context signal. This signal can then influence subsequent processing in the main STAC pathway (e.g., modulating attention or gating). This structure creates a form of associative memory where "the capacity of the memory is not determined by the width of the vectors, but instead, by the number of memory locations" Kleyko et al. (2021), or rather, the number of distinct patterns stored. This effective decoupling of representational capacity from vector dimensionality allows STAC to potentially maintain rich context information efficiently. Critically, this HDC approach enables the storage and retrieval of "ephemeral states"—temporary patterns of neural activity representing the current cognitive context or task-relevant information. These transient, dynamically updated memory states bear resemblance to the working memory reconfiguration observed during cognitive tasks in PFC function Miller and Cohen (2001), where specific neuronal populations maintain activity patterns representing goals or stimuli without requiring continuous external sensory input.

6.4 Phase Space Analysis

Bifurcation analysis, a tool from dynamical systems theory, provides a powerful mathematical framework for understanding how the qualitative behavior of STAC's spiking neurons (specifically, the AdEx model) transitions between different functional modes as key parameters (like input current I_{ext} or adaptation strength a) change. When examining the dynamics of AdEx neurons in the phase space (a plot of membrane potential V against the adaptation variable w), three distinct operational regimes typically emerge:

1. **Stable Focus (Quiescent):** For low input currents or high adaptation values, the neuron's state converges to a stable fixed point located below the firing threshold. In this regime, any small perturbations decay back to the resting state, creating an energy-efficient baseline where neurons remain inactive unless significant input arrives M. Wang and Mikhaeil (2018). Mathematically, this corresponds to a stable node or focus attractor in the phase space, with equilibrium values approximately at:

$$V_{eq} \approx V_{rest}$$
$$w_{eq} \approx 0$$

2. **Limit-Cycle (Persistent Memory):** As the input current increases beyond a critical bifurcation value (often driven by attention-related inputs in a cognitive model), the neuron can undergo a Hopf bifurcation. This transition leads from the stable fixed point to self-sustained, periodic oscillatory firing, represented by a stable limit cycle attractor in the phase space Naud and Gerstner (2012). This limit cycle enables neurons to maintain rhythmic activity even without continuous external driving input—providing the mathematical basis for persistent activity underlying working memory function. In this tonic firing regime, the firing

frequency f approximately follows:

$$f \approx \frac{1}{T_{ISI}} \propto \sqrt{I_{ext} - I_{critical}}$$

where T_{ISI} is the inter-spike interval and $I_{critical}$ is the critical current needed to initiate firing.

3. **Chaotic (Exploration):** Under certain parameter combinations, particularly intermediate adaptation values ($a \approx 4.0\,\text{nS}$, as used in STAC's PFC calibration) combined with strong or rapidly fluctuating input currents, AdEx neurons can exhibit irregular, non-periodic, or chaotic firing patterns Manjunath, Tino, and Jaeger (2012). While seemingly random, this regime can be computationally useful, potentially facilitating exploration of state space or enabling flexible switching between different response modes, which might be relevant for creative problem-solving or escaping local minima Sussillo and Barak (2013).

Bifurcation analysis by Naud and Gerstner Naud and Gerstner (2012) provides a mathematical foundation for STAC's operational regimes. Their derivation of Lyapunov exponents in AdEx neurons supports the chaotic exploration mode observed when $I_{ext} > 2.1\,\text{nA}$ Naud and Gerstner (2012). The transitions between these distinct dynamical regimes occur through specific, well-characterized bifurcations. For instance, subcritical or supercritical Hopf bifurcations typically mark the transition from quiescent to limit-cycle firing, while sequences of period-doubling bifurcations often precede the onset of chaotic dynamics as parameters are varied Gerstner, Kistler, Naud, and Paninski (2014). This mathematical analysis using phase space and bifurcation theory provides precise predictions about the conditions under which neurons will transit between different functional modes (e.g., resting, tonic firing, bursting, chaotic). These phase space analyses therefore complement SCAN's overall cognitively oriented framework by providing a detailed neurophysiological basis for different functional or

cognitive states potentially implemented by STAC. Quiescent modes represent energy conservation, limit-cycle dynamics support stable working memory representations, and chaotic or irregular dynamics might enable exploratory behavior or rapid state transitions—offering a mathematical foundation for the diverse cognitive functions SCAN aims to model.

7 Translational Perspectives

7.1 Neuroprosthetics and Executive Function

The real-time, low-power nature of spiking-based AI, as embodied in STAC, holds significant promise for neuroprosthetic devices aimed at assisting individuals with cognitive impairments. Such devices could benefit from STAC's capabilities by:

- Providing an External Memory Buffer: Temporarily storing task goals, multi-step instructions, or relevant context for individuals recovering from Traumatic Brain Injury (TBI) or stroke, who often exhibit deficits in working memory and executive control. The spiking module's ability to maintain activity (limit cycles) could represent these stored items.
- Enabling Adaptive Brain–Computer Interfaces (BCIs): Dynamically tuning the firing thresholds (V_{th}) or adaptation parameters (a, b) of the STAC SNN module based on real-time neural signals (e.g., EEG frequency bands associated with attention or cognitive load). This could allow the prosthetic to better align its assistance with the user's current cognitive state and intent.

This event-driven design, requiring computation only when relevant information arrives or needs updating, inherently favors lightweight, energy-efficient solutions suitable for wearable or implantable on-device deployment, offering the potential for continuous cognitive support throughout daily activities Davies et al. (2018).

7.2 AI Tutors and Adaptive Learning

Intelligent tutoring systems can significantly benefit from incorporating STAC's biologically inspired processing within the SCAN framework Tate (2024a). Potential applications include:

- Dynamic Lesson Pacing: Analyzing patterns of spike activity within STAC's SNN module (e.g., firing rates, synchrony, or transitions between firing modes) in response to learning materials. These patterns could potentially indicate student confusion (e.g., chaotic or overly sparse firing) or mastery (e.g., stable, predictable firing), allowing the tutor to adjust the pace or content accordingly.
- Maintaining Contextual Memory: Utilizing the hyperdimensional memory module or the persistent activity of AdEx neurons to retain ephemeral knowledge states related to the student's recent performance, misconceptions, or correctly acquired concepts. This allows the AI tutor to provide more individualized instruction and feedback that builds upon the student's unique learning trajectory in real time.

This combination of event-driven efficiency and cognitive modeling fosters the development of truly adaptive learning systems capable of real-time personalization with minimal computational overhead, effectively bridging neuromorphic principles with user-centric AI design.

8 Critical Gaps and Mitigation Strategies

8.1 Addressing Limited Quantitative Benchmarks

A key challenge is the relative scarcity of direct quantitative comparisons on standardized cognitive benchmarks. While STAC achieves a significant theoretical reduction in computational operations (estimated at 72.75% via sparse matrix multiplications

based on average activity, calculated as $1 - (0.35)^2 - 0.15$, where 0.35 represents average spiking activity and 0.15 accounts for encoding overhead), empirical validation on tasks demanding working memory, gating, or task-switching is needed.

- **Mitigation:** Future work should focus on evaluating STAC on established cognitive neuroscience benchmark suites (e.g., modified Stroop tasks, N-back tasks, sequence learning paradigms) adapted for sequential models. This will allow for direct comparison of performance (accuracy, reaction time analogs) and efficiency (FLOPs, simulated energy) against both standard Transformers and other SNN–Transformer hybrids. The calculated efficiency makes STAC particularly suitable for real-time applications requiring immediate responses, resource-constrained environments like edge computing, while its integration within the SCAN ecosystem enhances its potential for cognitively rich tasks in neuroprosthetics, adaptive learning, and human–AI collaboration contexts.

8.2 Comparisons With State-of-the-Art Models

Direct, head-to-head evaluations against other leading SNN–Transformer architectures like Spikformer Zhou, Li, Chen, and He (2022) or SpikeGPT Zhu, Li, and Tang (2023) are currently limited. Furthermore, STAC's explicit emphasis on PFC-inspired cognitive tasks (working memory, gating) contrasts with many existing models that are often optimized for purely classification (e.g., image recognition Shi et al. (2024), medical imaging Xie et al. (2024)) or generative language tasks.

- **Mitigation:** Implementing standardized benchmark protocols that encompass both traditional ML tasks (to assess general capabilities) and specialized cognitive tasks is crucial. This would help elucidate the specific trade-offs in efficiency, accuracy, and domain generality offered by STAC's architecture compared to alternatives. Testing on diverse datasets beyond WikiText-2 is also necessary.

8.3 Deepening Neuroscience Alignment

While AdEx neurons calibrated with DLPFC data C. Wang et al. (2013) provide a degree of biological plausibility, the current model represents a simplification of complex cortical circuits. It lacks representations of distinct neuron types (inhibitory interneurons), glial cell interactions (like astrocytes, known to modulate synaptic plasticity and neuronal firing Nourianzadeh and Valizadeh (2021)), or the effects of crucial neuromodulators like dopamine, which heavily influences PFC function and reward-based learning Schultz (2016).

- **Mitigation:** Incrementally incorporating more detailed biological features, such as including populations of inhibitory neurons, modeling astrocyte–neuron interactions Nourianzadeh and Valizadeh (2021), or integrating dopaminergic gating mechanisms Schultz (2016) linked to reinforcement learning signals, could significantly enhance the model's biological realism and potentially its functional capabilities for complex cognitive tasks.

8.4 Hardware Co-Design for Neuromorphic Deployment

Claims regarding STAC's energy efficiency are currently based on theoretical calculations (FLOP reduction) and simulations. True validation requires deployment and testing on actual neuromorphic hardware platforms like Intel's Loihi 2 Davies et al. (2018) or BrainChip's Akida. Furthermore, relying solely on surrogate gradient methods Shrestha and Orchard (2018) for training SNNs might not be optimal for on-chip learning in future hardware.

- **Mitigation:** Actively collaborating with neuromorphic hardware developers to port and test STAC components is essential. This would provide direct, empirical data on power consumption and performance. Exploring the integration of on-chip plasticity mechanisms, such as Spike-Timing-Dependent Plasticity (STDP), could potentially

supplant or augment surrogate backpropagation Shrestha and Orchard (2018) for learning directly on neuromorphic chips in production deployments.

8.5 Ensuring Scalability

While the spiking module's parameter overhead is designed to be relatively low ($\sim$12%) and constant across different base Transformer sizes Rokh et al. (2022), ensuring efficient training and inference for extremely large models (e.g., 100B+ parameters) remains a challenge. Memory usage, particularly for activations and optimizer states during training, could still be substantial.

- **Mitigation:** Investigating the application of techniques like parameter-efficient fine-tuning (PEFT) methods (e.g., LoRA, Adapters) specifically adapted for the hybrid SNN–Transformer architecture could significantly reduce memory footprint and training costs. Exploring model parallelism strategies optimized for the sparse, event-driven nature of the SNN component might also yield further scalability improvements for deploying STAC with extremely large LLMs.

8.6 Ethical and Societal Considerations

The development of efficient, potentially pervasive event-driven AI like STAC raises important ethical questions regarding privacy (continuous monitoring potential), algorithmic bias amplification, and energy justice. While low-power design broadens AI accessibility in resource-limited locales, it could also exacerbate disparities if not deployed equitably.

- **Mitigation:** Adopting a proactive stance on AI ethics is crucial. This includes designing for transparency, conducting bias audits, prioritizing human–AI collaboration frameworks (as facilitated by SCAN) rather than aiming for full autonomy, and engaging in open dialogue about the societal implications and equitable deployment of low-power AI technologies in diverse global contexts. Furthermore, addressing the

ethical landscape requires engaging with emerging frameworks specifically tailored to neuromorphic and cognitive augmentation technologies. For instance, Crawford Crawford (2024) emphasizes that low-power AI systems like STAC must address energy justice considerations proactively to avoid exacerbating global inequities in access to computational resources and their benefits. Concurrently, the rapid development of brain-computer interfaces, closely related to neuroprosthetics discussed earlier, necessitates robust ethical guidelines. Yuste et al. Yuste et al. (2024) propose such guidelines, advocating particularly for the use of techniques like differential privacy in spike-based BCIs to rigorously protect sensitive neural data and preserve mental privacy for users of cognitive augmentation technologies Crawford (2024); Yuste et al. (2024). Integrating these considerations into STAC's design and deployment strategy will be vital for responsible innovation.

9 Detailed Analysis of Hybrid SNN–Transformer Architectures

9.1 Neuromorphic Computing and Spiking Neural Networks (SNNs)

Neuromorphic computing paradigms draw inspiration from the brain's structure and function, particularly its event-driven nature. Unlike traditional synchronous systems, neuromorphic designs often trigger computations only when a significant event, such as a neuron's membrane potential crossing a threshold, occurs Brette and Gerstner (2005). This asynchronous, sparse activity pattern is well-suited for real-time processing and tasks where inputs are inherently sparse or bursty, offering substantial energy savings compared to continuously active systems. STAC's SNN gating

paradigm directly aligns with these principles, leveraging threshold-based AdEx neurons to favor computational concurrency with minimal power consumption during periods of low input activity Intel Corporation (2021).

9.2 Transformer Architectures in Artificial Intelligence

Transformer models Vaswani et al. (2017) have become the dominant architecture for large-scale sequence modeling tasks, primarily due to their powerful self-attention mechanism that captures long-range dependencies in data. However, this capability comes at the cost of high computational and memory demands, particularly the quadratic complexity of self-attention with respect to sequence length Dosovitskiy et al. (2020). STAC's hybrid approach aims to mitigate this by using the spiking modules to selectively gate information flow, theoretically reducing the effective number of tokens processed by the attention mechanism by filtering out less salient information based on threshold crossings. This seeks to balance the Transformer's capacity for capturing broad context with the efficiency gains of event-driven SNN processing.

9.3 Hybrid Integration Methodologies

Various strategies exist for integrating SNNs and Transformers. Many current systems focus on replacing specific components of a standard Transformer with spiking equivalents or using SNNs as input encoders, often targeting classification Chen et al. (2023); Guo et al. (2024); Liu, Guo, and Zheng (2024); Shi et al. (2024) or generative tasks Zhou et al. (2022); Zhu et al. (2023). STAC distinguishes itself by focusing explicitly on implementing higher-level cognitive functions inspired by the PFC, such as working memory maintenance (via persistent firing), top-down attentional control (via spike-based gating modulating Transformer attention), and enabling multi-agent synergy within the cognitive architecture framework of SCAN Tate (2024a); Tate and Sanders (2024b). The integration aims to achieve functional biomimicry beyond simple component replacement.

9.4 Advancements in Neuromorphic Hardware

The field of neuromorphic hardware is rapidly evolving, with platforms like Intel Loihi 2 Davies et al. (2018), IBM TrueNorth, SpiNNaker, and BrainChip Akida demonstrating increasing capabilities for implementing SNNs directly in silicon Intel Corporation (2021). These chips often feature massively parallel cores, asynchronous communication, and sometimes on-chip learning capabilities (e.g., STDP). STAC's event-driven gating mechanism is conceptually well-suited for implementation on such spiking cores. Future integration could leverage hardware-specific features like local plasticity rules and low-latency, asynchronous processing to manage large-scale cognitive simulations with potentially orders-of-magnitude lower power consumption than conventional hardware.

9.5 Real-World Deployment and Future Implications

The unique characteristics of hybrid SNN–Transformer models like STAC open up possibilities in several application domains:

- **Edge Computing:** The low idle power consumption of spiking modules makes them ideal for battery-limited edge devices (wearables, sensors, mobile phones) that require intermittent but powerful sequence processing capabilities.
- **Robotics:** Real-time gating based on spike thresholds can allow robots to efficiently filter potentially massive streams of sensor data, focusing processing resources only on relevant or unexpected events, enabling faster reaction times and more efficient navigation or interaction.
- **Cognitive Augmentation:** Integration within frameworks like SCAN facilitates user-centric applications, ranging from personalized AI tutors that adapt to cognitive states Tate (2024a) to real-time decision support systems or assistive technologies for individuals with cognitive impairments Davies et al. (2018).

9.6 Overall Evaluation and Summary of the Detailed Analysis

STAC represents a deliberate effort to merge the computational efficiency and biological plausibility of SNNs with the powerful contextual modeling capabilities of Transformers, specifically targeting cognitive functions associated with the PFC. By embedding this hybrid architecture within the SCAN multi-agent ecosystem, it aims to create systems capable of more complex, human-like cognitive tasks than typical classification-focused models. While challenges remain, particularly in empirical validation and hardware deployment, the approach offers a promising pathway toward more efficient, scalable, and cognitively sophisticated AI. Further developments—including rigorous benchmarking, deeper neuroscience integration (e.g., neuromodulatory modeling), and co-design with neuromorphic hardware—will be crucial to fully realize STAC's potential energy and functional advantages in real-world deployments.

10 Conclusion: A New Paradigm for Biologically Inspired AI

By strategically joining principles of event-driven, spiking neuromorphic design with the representational power of Transformer-based sequence modeling, STAC offers a novel computational framework aimed at capturing key Prefrontal Cortex functionalities, notably working memory and attentional gating. This focus distinguishes it from many existing spiking–transformer hybrids that currently revolve around more specialized, often sensory-based tasks such as defect classification in manufacturing Guo et al. (2024) or object recognition in satellite imagery Xu, Jia, Lu, and Qi (2024). STAC, in contrast, targets a broader cognitive domain and is explicitly positioned within the SCAN multi-agent ecosystem Tate and Sanders (2024b) to facilitate complex, collaborative cognitive processes. The careful calibration of its AdEx neurons

using parameters inspired by experimental DLPFC data Miller and Cohen (2001); C. Wang et al. (2013) provides a degree of biological grounding, while the modular design ensures that the SNN component adds only a modest computational overhead (estimated theoretically at ∼12% in parameters Rokh et al. (2022)), allowing scalability to large Transformer backbones without incurring prohibitive costs. The anticipated synergy of spike-based gating is theoretically projected to yield substantial computational savings (e.g., ∼72% FLOPs reduction based on simulated average spike rates of ∼35%), although empirical validation is pending. Such minimal operational overhead, particularly during periods of low input activity, makes the architecture well-suited for real-time or resource-constrained environments, effectively bridging modern machine learning techniques with core principles from computational neuroscience to create cognitively enriched AI systems. Future endeavors should prioritize deepening the neuroscientific fidelity—perhaps through the incorporation of astrocyte modeling for synaptic modulation Nourianzadeh and Valizadeh (2021) or dopaminergic mechanisms for reinforcement learning Schultz (2016)—and critically, validating the theoretical energy efficiency claims through rigorous testing on dedicated neuromorphic hardware platforms Davies et al. (2018). STAC's event-driven, cognition-inspired hybrid approach opens promising avenues for developing more capable and efficient AI, potentially enabling more equitable deployment in power-limited settings and fostering human–AI collaboration–based solutions rather than pursuing fully autonomous systems. Lastly, the broader impact of STAC's success will resonate in discussions about the future of AI hardware-software co-design. As neuromorphic processors mature, featuring spikes as their intrinsic language of computation, software frameworks like STAC are poised to become essential. They will likely guide best practices in designing algorithms that operate asynchronously, reduce redundancy, and align with the constraints and opportunities offered by spiking-based hardware

Intel Corporation (2021). Within large-scale deployments, such as data centers dedicated to running NLP services, the event-driven approach could significantly alleviate the energy burden traditionally associated with Transformers. Similarly, resource-constrained settings—disaster zones requiring low-power devices, remote locations with limited infrastructure, or wearable sensors for continuous monitoring—stand to benefit immensely. By establishing a synergy between the SCAN multi-agent ecosystem, spiking-based gating, and advanced attention mechanisms, STAC paves the way for not only more sustainable AI but also for a deeper conceptual understanding of how biologically inspired design principles can transform mainstream machine learning. If these research directions materialize, they could ultimately usher in an era where AI systems grow increasingly adept at seamlessly integrating with human cognitive processes, offering assistance in real time, with minimal resource overhead and a heightened respect for privacy and ethical deployment.

Supplementary information. Not applicable.

Declarations

- Funding: The authors did not receive support from any organization for the submitted work.
- Competing interests: The authors have no relevant financial or non-financial interests to disclose.
- Ethics approval: Not applicable.
- Consent to participate: Not applicable.
- Consent for publication: Not applicable.
- Data availability: The datasets generated during and/or analysed during the current study are available in the Zenodo repository, Tate (2025).
- Code availability: The code generated during the current study is available in the Zenodo repository, Tate (2025).

- Materials availability: Not applicable.
- Author contribution: B.J.K. conducted the research, developed the software/model, and wrote the initial draft of the manuscript. A.F.M. and M.S.W. supervised the project as dissertation chairs and provided critical review and editing. All authors read and approved the final manuscript.

References

Baars, B.J. (2002). The conscious access hypothesis: Origins and recent evidence. *Trends in Cognitive Sciences*, *6*(1), 47–52, https://doi.org/10.1016/S1364-6613(00)01819-2 Retrieved from https://doi.org/10.1016/S1364-6613(00)01819-2

Brette, R., & Gerstner, W. (2005). Adaptive exponential integrate-and-fire model as an effective description of neuronal activity. *Journal of Neurophysiology*, *94*(5), 3637–3642, https://doi.org/10.1152/jn.00184.2005 Retrieved from https://doi.org/10.1152/jn.00184.2005

Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J.D., Dhariwal, P., ... Amodei, D. (2020). Language models are few-shot learners. H. Larochelle, M. Ranzato, R. Hadsell, M.F. Balcan, & H. Lin (Eds.), *Advances in neural information processing systems 33* (pp. 1877–1901). Curran Associates, Inc. Retrieved from https://proceedings.neurips.cc/paper/2020/hash/1457c0d6bfcb4967418bfb8ac142f64a-Abstract.html

Buonomano, D.V., & Maass, W. (2009). State-dependent computations: Spatiotemporal processing in cortical networks. *Nature Reviews Neuroscience*, *10*(2), 113–125, https://doi.org/10.1038/nrn2558 Retrieved from

https://doi.org/10.1038/nrn2558

Chen, R., Ma, H., Liu, S., Liu, T., Tang, M. (2023). *Enhancing the performance of transformer-based spiking neural networks by SNN-optimized downsampling with precise gradient backpropagation.* arXiv preprint arXiv:2305.05954. Retrieved from https://arxiv.org/abs/2305.05954

Chowdhury, S.N., & Ermentrout, G.B. (2024). Bifurcation control in adaptive exponential integrate-and-fire networks. *SIAM Journal on Applied Dynamical Systems*, *23*(1), 1–34, https://doi.org/10.1137/23M1567123 Retrieved from https://doi.org/10.1137/23M1567123

Crawford, K. (2024). Energy justice in low-power AI: A framework for equitable neuromorphic deployment. *Science, Technology, & Human Values*, *49*(2), 345–372, https://doi.org/10.1177/01622439231177287 Retrieved from https://doi.org/10.1177/01622439231177287

Davies, M., Srinivasa, N., Lin, T.-H., Chinya, G., Cao, Y., Choday, S.H., ... Narayan, A. (2018). Loihi: A neuromorphic manycore processor with on-chip learning. *IEEE Micro*, *38*(1), 82–99, https://doi.org/10.1109/MM.2018.1121303 Retrieved from https://doi.org/10.1109/MM.2018.1121303

Dehaene, S., Lau, H., Kouider, S. (2017). What is consciousness, and could machines have it? *Science*, *358*(6362), 486–492, https://doi.org/10.1126/science.aan8871 Retrieved from https://doi.org/10.1126/science.aan8871

Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., ... Houlsby, N. (2020). *An image is worth 16x16 words: Transformers for image recognition at scale.* arXiv preprint arXiv:2010.11929. Retrieved from https://doi.org/10.48550/arXiv.2010.11929

Gerstner, W., Kistler, W.M., Naud, R., Paninski, L. (2014). *Neuronal dynamics: From single neurons to networks and models of cognition.* Cambridge University Press. Retrieved from https://doi.org/10.1017/CBO9781107447615

Guo, L., Zhang, D., Wang, L., Zhou, H., Chen, Y. (2024). *Transformer-based spiking neural networks for multimodal audio-visual classification.* Semantic Scholar. Retrieved from https://www.semanticscholar.org/paper/Transformer-based-Spiking-Neural-Networks-for-Guo-Zhang/8992e9ee1c0ee070b2d5ea2cd15e3ac071bba6 (Paper ID 8992e9ee1c0ee070b2d5ea2cd15e3ac071bba6)

Indiveri, G., & Liu, S.-C. (2015). Memory and information processing in neuromorphic systems. *Proceedings of the IEEE*, *103*(8), 1379–1397, https://doi.org/10.1109/JPROC.2015.2444094 Retrieved from https://doi.org/10.1109/JPROC.2015.2444094

Intel Corporation (2021). *Taking neuromorphic computing with Loihi 2 to the next level: Technology brief* (Tech. Rep.). Intel Corporation. Retrieved from https://download.intel.com/newsroom/2021/new-technologies/neuromorphic-computing-lohi-2-brief.pdf

Kanerva, P. (2009). Hyperdimensional computing: An introduction to computing in distributed representation with high-dimensional random vectors. *Cognitive Computation*, *1*(2), 139–159, https://doi.org/10.1007/s12559-009-9009-8

Retrieved from https://doi.org/10.1007/s12559-009-9009-8

Kleyko, D., Rachkovskij, D.A., Osipov, E., Rahimi, A. (2021). Vector symbolic architectures as a computing framework for nanoscale hardware. *IBM Journal of Research and Development*, *65*(4/5), 3:1–3:15, https://doi.org/10.1147/JRD.2021.3064773 Retrieved from https://doi.org/10.1147/JRD.2021.3064773

Liu, W., Guo, L., Zheng, X. (2024). *Optimizing spiking neural networks with transformer-based spatial attention.* Semantic Scholar. Retrieved from https://www.semanticscholar.org/paper/Optimizing-Spiking-Neural-Networks-with-Spatial-Liu-Guo/e946a6f7301371176a24ed9894a350f40ee26172 (Paper ID e946a6f7301371176a24ed9894a350f40ee26172)

Manjunath, G., Tino, P., Jaeger, H. (2012). Theory of input driven dynamical systems. *Proceedings of the european symposium on artificial neural networks, computational intelligence and machine learning (esann)* (pp. 25–27). Bruges, Belgium.

Miller, E.K., & Cohen, J.D. (2001). An integrative theory of prefrontal cortex function. *Annual Review of Neuroscience*, *24*(1), 167–202, https://doi.org/10.1146/annurev.neuro.24.1.167 Retrieved from https://doi.org/10.1146/annurev.neuro.24.1.167

Naud, R., & Gerstner, W. (2012). Coding and decoding with adapting neurons: A population approach to the peri-stimulus time histogram. *PLoS Computational Biology*, *8*(10), e1002711, https://doi.org/10.1371/journal.pcbi.1002711 Retrieved from https://doi.org/10.1371/journal.pcbi.1002711

Nourianzadeh, Y., & Valizadeh, A. (2021). Sustained activity in a spiking neuron model with astrocytes: The role of tripartite synapses in working memory tasks. *PLoS Computational Biology*, *17*(5), e1009012, https://doi.org/10.1371/journal.pcbi.1009012 Retrieved from https://doi.org/10.1371/journal.pcbi.1009012

Plate, T.A. (2003). *Holographic reduced representation: Distributed representation for cognitive structures.* Stanford, CA: CSLI Publications.

PyTorch Team (2020). *PyTorch: An open source machine learning framework.* Software. Retrieved from https://pytorch.org/

Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., Sutskever, I. (2019). *Language models are unsupervised multitask learners.* OpenAI Blog. Retrieved from https://cdn.openai.com/better-language-models/language$_m odels_a re_u nsupervised_m ultitask_l earners.pdf$ (Version 1.8, page 9)

Rokh, B., Azarpeyvand, A., Khanteymoori, A. (2022). A comprehensive survey on model quantization for deep neural networks in image classification. *ACM Transactions on Intelligent Systems and Technology*, *14*(6), 1–50, https://doi.org/10.1145/3565104 Retrieved from https://doi.org/10.1145/3565104

Schultz, W. (2016). Reward functions of the basal ganglia. *Journal of Neural Transmission*, *123*(7), 679–693, https://doi.org/10.1007/s00702-016-1527-9 Retrieved from https://doi.org/10.1007/s00702-016-1527-9

Shi, X., Hao, Y., Wang, L. (2024). *SpikingResformer: Bridging ResNet and vision transformer in spiking neural networks.* arXiv preprint arXiv:2403.14302. Retrieved from https://arxiv.org/abs/2403.14302

Shrestha, S.B., & Orchard, G. (2018). Slayer: Spike layer error reassignment in time. S. Bengio, H. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, & R. Garnett (Eds.), *Advances in neural information processing systems 31* (pp. 1419–1428). Curran Associates, Inc. Retrieved from https://proceedings.neurips.cc/paper/2018/hash/82f2d5c9c73c5a35aa9ed11afcce4ca0-Abstract.html

Sussillo, D., & Barak, O. (2013). Opening the black box: Low-dimensional dynamics in high-dimensional recurrent neural networks. *Neural Computation*, *25*(3), 626–649, https://doi.org/10.1162/NECO_a_00409 Retrieved from https://doi.org/10.1162/NECO$_{a0}$0409

Tate, L. (2024a). *iLevyTate/SCANUE: 1.0.0-alpha.* Computer software. Zenodo. Retrieved from https://doi.org/10.5281/zenodo.14052759

Tate, L. (2025). *iLevyTate/stac: 1.0.2.2-alpha.* Computer software. Zenodo. Retrieved from https://doi.org/10.5281/zenodo.15238162

Tate, L., Mohammad, A.F., Wyandt, M.S., Kennedy, B.J., Sanders, P., Blackwood, A., Levy, J. (2024c). *SCAN resources.* Computer software. Zenodo. Retrieved from https://doi.org/10.5281/zenodo.14053203

Tate, L., & Sanders, P. (2024b). *iLevyTate/SCAN: 1.0.0-alpha.* Computer software. Zenodo. Retrieved from https://doi.org/10.5281/zenodo.14052885

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., ... Polosukhin, I. (2017). Attention is all you need. I. Guyon et al. (Eds.), *Advances in neural information processing systems 30* (pp. 5998–6008). Curran Associates, Inc. Retrieved from https://papers.nips.cc/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html

Wang, C., Armstrong, K.M., Jeong, H.-Y., Cornelisse, L.N., Jo, J., Arnsten, A.F.T., Wang, M. (2013). NMDA receptors subserve persistent neuronal firing during working memory in dorsolateral prefrontal cortex. *Proceedings of the National Academy of Sciences of the United States of America*, *110*(48), 19436–19441, https://doi.org/10.1073/pnas.1315048110 Retrieved from https://doi.org/10.1073/pnas.1315048110

Wang, M., & Mikhaeil, J.L. (2018). Bifurcation analysis of adaptive exponential integrate-and-fire neurons. *Biological Cybernetics*, *112*(5), 483–497, https://doi.org/10.1007/s00422-018-0768-0 Retrieved from https://doi.org/10.1007/s00422-018-0768-0

Xie, M., Ding, Y., Yang, D., Shen, J., Zhu, J., Ma, Q. (2024). *EG-SpikeFormer: Eye-gaze guided transformer on spiking neural networks for medical image analysis.* arXiv preprint arXiv:2410.09674. Retrieved from https://arxiv.org/abs/2410.09674

Xu, W., Jia, H., Lu, C., Qi, Y. (2024). *RepSViT: An efficient vision transformer based on spiking neural networks for object recognition in satellite on-orbit remote sensing images.* Semantic Scholar.

Retrieved from https://www.semanticscholar.org/paper/RepSViT (Paper ID 771222df1319dbc712efa32184ac902c78141b71)

Yuste, R., Goering, S., Bi, G., Carmena, J.M., Carter, A., Fins, J.J., ... Wolpaw, J. (2024). Ethical guidelines for neuroprosthetics and cognitive augmentation. *Nature Machine Intelligence*, *6*(4), 321–329, https://doi.org/10.1038/s42256-024-00812-y Retrieved from https://doi.org/10.1038/s42256-024-00812-y

Zhou, Z., Li, Y., Chen, Y., He, Z. (2022). *Spikformer: When spiking neural network meets transformer.* ICLR 2023 Conference Paper. Retrieved from https://openreview.net/forum?id=frE4fUwz$_h$

Zhu, S., Li, J., Tang, S. (2023). *SpikeGPT: Generative pre-trained language model with spiking neural networks.* arXiv preprint arXiv:2302.13939. Retrieved from https://arxiv.org/abs/2302.13939